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import matplotlib.pyplot as plt

# Load and preprocess the MNIST dataset

(x_train, y_train), (x_test, y_test) = datasets.mnist.load_data()

x_train, x_test = x_train / 255.0, x_test / 255.0 # Normalize pixel values to [0, 1]


# Reshape data to fit the CNN (add a channel dimension)

x_train = x_train.reshape(-1, 28, 28, 1)

x_test = x_test.reshape(-1, 28, 28, 1)


# Build the CNN model

model = models.Sequential([

    layers.Conv2D(32, (3, 3), activation='relu', input_shape=(28, 28, 1)),

    layers.MaxPooling2D((2, 2)),

    layers.Conv2D(64, (3, 3), activation='relu'),

    layers.MaxPooling2D((2, 2)),

    layers.Conv2D(64, (3, 3), activation='relu'),

    layers.Flatten(),

    layers.Dense(64, activation='relu'),

    layers.Dense(10, activation='softmax') # 10 classes for MNIST digits 0-9

])


# Compile the model

model.compile(optimizer='adam',

              loss='sparse_categorical_crossentropy',

              metrics=['accuracy'])
```

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```
# Train the model

history = model.fit(x_train, y_train, epochs=5, validation_data=(x_test, y_test))

# Evaluate the model

test_loss, test_acc = model.evaluate(x_test, y_test, verbose=2)

print(f'\nTest accuracy: {test_acc}')

# Plot training and validation accuracy over epochs

plt.plot(history.history['accuracy'], label='accuracy')

plt.plot(history.history['val_accuracy'], label = 'val_accuracy')

plt.xlabel('Epoch')

plt.ylabel('Accuracy')

plt.legend(loc='lower right')

plt.show()
```

OUTPUT

```
Epoch 1/5

1875/1875 ————— 24s 12ms/step - accuracy: 0.9002 -
loss: 0.3191 - val_accuracy: 0.9826 - val_loss: 0.0513

Epoch 2/5

1875/1875 ————— 21s 11ms/step - accuracy: 0.9849 -
loss: 0.0480 - val_accuracy: 0.9876 - val_loss: 0.0360

Epoch 3/5

1875/1875 ————— 23s 12ms/step - accuracy: 0.9903 -
loss: 0.0330 - val_accuracy: 0.9881 - val_loss: 0.0389

Epoch 4/5
```

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1875/1875 ————— 23s 12ms/step - accuracy: 0.9925 -
loss: 0.0238 - val_accuracy: 0.9871 - val_loss: 0.0400

Epoch 5/5

1875/1875 ————— 24s 13ms/step - accuracy: 0.9944 -
loss: 0.0187 - val_accuracy: 0.9921 - val_loss: 0.0283

313/313 - 2s - 5ms/step - accuracy: 0.9921 - loss: 0.0283

Test accuracy: 0.9921000003814697

